

Sowmya Yalavarthi

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Professional Summary

Data and Analytics professional with around 5 years of experience designing performance measurement frameworks, KPI dashboards, and data pipelines that improve decision-making across large organizations. Skilled at blending quantitative analysis with qualitative insights to monitor quality outcomes, track safety trends, and benchmark performance against internal and external standards. Adept at translating complex datasets into actionable recommendations for leadership, enabling faster decisions, improved compliance, and measurable operational gains. Experienced in SQL, Tableau, and Python, with a strong background in healthcare analytics, reporting automation, and stakeholder collaboration.

Technical Skills

Data & Analytics: SQL (Snowflake, PostgreSQL, SQL Server), Data Profiling, Data Mapping, KPI Reporting, Benchmarking

ETL & Warehousing: Airflow, ETL/ELT pipelines, Data Wrangling, Snowflake

Visualization & BI: Tableau, Power BI, Performance Dashboards, KPI Tracking

Programming: Python (Pandas, NumPy), Excel (Pivot Tables, VLOOKUPS)

Cloud Platforms: AWS (S3, Glue, RDS, CloudWatch), GCP (BigQuery, Pub/Sub)

Collaboration: Agile (Scrum/Kanban), Stakeholder Engagement, Executive Reporting

Experience

Data Analyst

Jan 2025 – Present

Tern

New York, NY

- Partnered with operations and compliance leaders to design performance dashboards that tracked outcome-based KPIs, such as retention, response times, and engagement across critical workflows. These dashboards standardized executive reviews and improved transparency into areas needing corrective action.
- Automated quality monitoring pipelines using Airflow and Snowflake to aggregate patient engagement and service delivery data from multiple systems. This reduced manual prep time by 35% and enabled leadership to review real-time performance against benchmarks.
- Conducted root-cause analyses of workflow delays by integrating survey feedback with quantitative data, uncovering gaps in patient onboarding processes. Insights led to target interventions that improved service cycle times by 12%.
- Created A/B testing frameworks in Python to evaluate the impact of new patient communication protocols, allowing the organization to adopt data-backed methods that increased patient satisfaction scores.
- Documented reporting workflows and mentored junior analysts on SQL optimization and dashboard design, ensuring consistency and accuracy in operational reporting.

Data Analyst

June 2023 – Sept 2023

Tata Consultancy Services

Hyderabad, India

- Migrated operational reporting workloads from SQL Server to Amazon RDS for PostgreSQL, redesigning schemas and optimizing queries to reduce refresh times by 25% while maintaining compliance with internal governance standards.
- Developed Snowflake SQL procedures to support patient outcome dashboards, enabling clinicians and administrators to track safety events and process adherence in near real time.
- Built Power BI dashboards to monitor KPIs such as error rates, throughput, and backlog. These dashboards enabled quicker intervention from managers, reducing bottlenecks in service delivery.
- Designed forecasting models in Python and SQL to simulate patient demand trends, improving workforce planning accuracy by 12% and helping leadership allocate resources more effectively.
- Implemented anomaly detection logic to flag irregular patterns in performance data, allowing corrective actions to be taken before issues escalated into compliance risks.
- Led cross-functional sessions to align KPI definitions with organizational goals, ensuring leadership reports reflect consistent and meaningful benchmarks across teams.

Data Analyst

Dec 2019 – May 2023

Capgemini

Hyderabad, India

- Built automated ETL pipelines to ingest data from Oracle into Snowflake, enabling daily performance reporting at scale while ensuring adherence to SLA and compliance requirements.
- Developed Power BI dashboards for clinical and operational leaders to monitor KPIs such as patient

throughput, service times, and quality outcomes, increasing transparency and accountability across departments.

- Optimized Snowflake SQL queries and implemented clustering strategies that cut report refresh times by 30%, making daily operational dashboards accessible to leadership faster.
- Designed standardized templates in Python and SQL for analyzing intervention outcomes, enabling teams to measure effectiveness with statistical rigor.
- Partnered with operations and compliance teams to align organizational priorities with measurable performance indicators, ensuring data-driven improvements in quality and patient safety.
- Built anomaly detection scripts to identify irregular patient flow and reporting errors, reducing wasted effort and increasing trust in the analytics function.
- Introduced automated Airflow-based workflows to enforce consistent reporting standards across quality teams, increasing efficiency and reducing the margin of error in recurring reports.

Projects

Climate-Aware Crop Yield Forecasting	Jan 2025 – May 2025
• Trained Random Forest and XGBoost models on 50K+ global climate records, achieving $R^2 = 0.89$ for predicting agricultural outcomes under varying conditions. • Applied SHAP explainability to highlight key climate risk drivers (rainfall, CO₂, temperature). • Built a Streamlit app integrating geospatial datasets, allowing stakeholders to simulate climate-risk scenarios and visualize yield outcomes.	
AI for Healthcare and Wellbeing	Sept 2024 – Dec 2024
• Developed an AI chatbot for mental health check-ins using Gemini API and Hugging Face NLP models, applying sentiment analysis and journaling prompts to improve user engagement. • Built a Streamlit UI with cohort analysis and A/B testing, increasing user retention by 30% during pilot trials. • Earned recognition at a Google Hackathon for the chatbot’s innovation and accessibility in risk-sensitive user contexts.	

Education

Pace University	New York
<i>Master’s in data science</i>	<i>Sept. 2023 – May 2025</i>
KL University	India
<i>Bachelor of Technology in Electronics and Communication Engineering</i>	<i>June 2015 – May 2019</i>

Certifications And Achievements

- AWS Cloud Practitioner (CLF-C02)
- Capgemini Customer Delight Award
- Google Hackathon (2025)
- Microsoft Power BI